

HAT-P-36b

R-1301

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_210316 · created 2026-07-31T17:18:08+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459289.7182 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.011
Transit depth (Rp/Rs)^2	2.2 +/- 0.31 [%]
Orbital Inclination (inc)	83.5 +/- 2.2
Airmass coefficient 1 (a1)	1.263 +/- 0.016
Airmass coefficient 2 (a2)	-0.1959 +/- 0.0066
Scatter in the residuals of the lightcurve fit is	1.28 %
Best Comparison Star	None
Optimal Aperture	4.39
Optimal Annulus	7.32
Transit Duration (day)	0.0884 +/- 0.0078

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.148 $\pm$ 0.011 vs 0.126 (deviation 2.0 σ)
Duration fit vs published	0.0884 d vs 0.0925 d (deviation 0.53 σ)
Mid-time O–C	1.7 min
Depth SNR	13.5
Residual scatter	1.28 %

Light curve

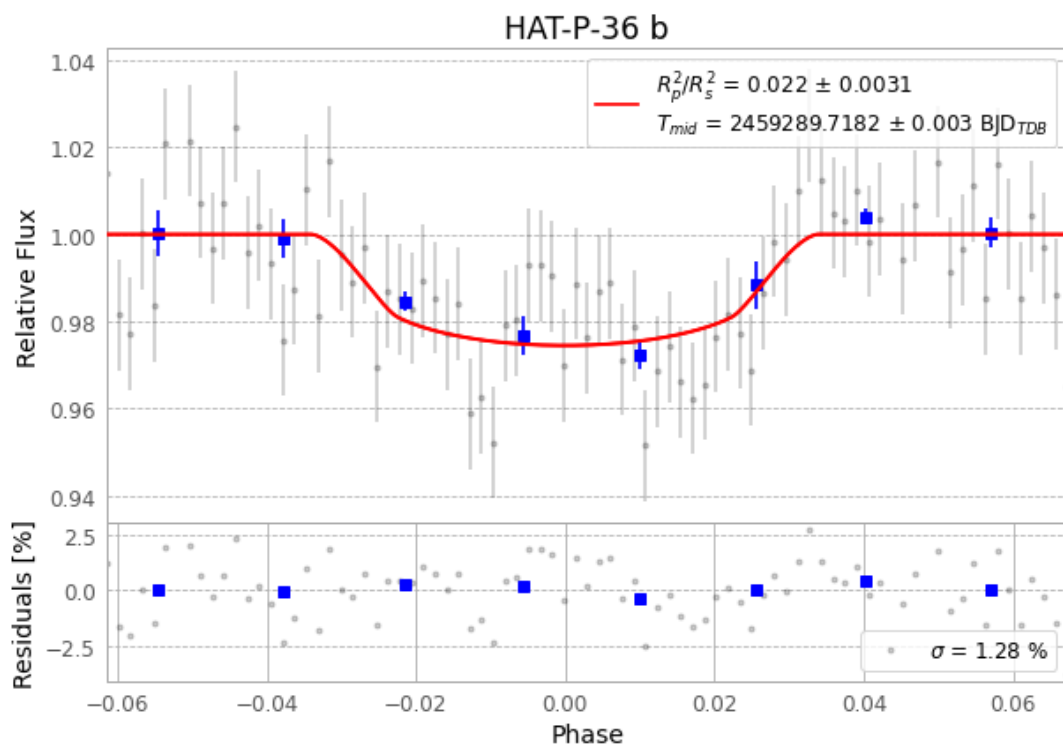


Figure 1 — FinalLightCurve_HAT-P-36 b_2021-03-15.png (SHA-256 cea53f6d975f2cee...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [319, 380], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 51a32a00155c3847...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), HATP-36210316030912.FITS → HATP-36210316071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-210316.log (3af16d4788fe...), FinalLightCurve_HAT-P-36 b_2021-03-15.png (cea53f6d975f...), FinalLightCurve_HAT-P-36 b_2021-03-15.csv (c0ef4ea2b313...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 17:18Z.